Q1. Nikki tries to set up four electric circuits.

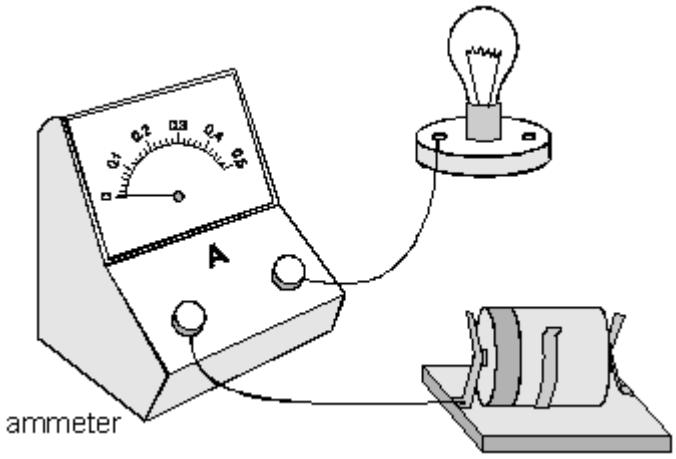


diagram 1

(a) In diagram 1 the ammeter reading is zero. What is wrong with this circuit?

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1 mark

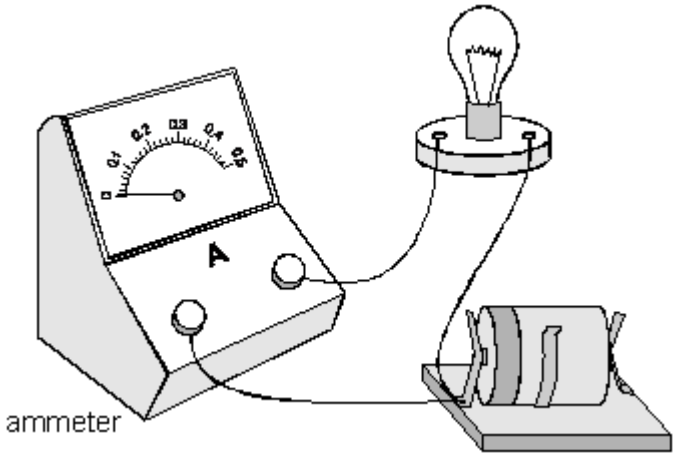


diagram 2

(b) In diagram 2 the ammeter reading is zero. What is wrong with this circuit?

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1 mark

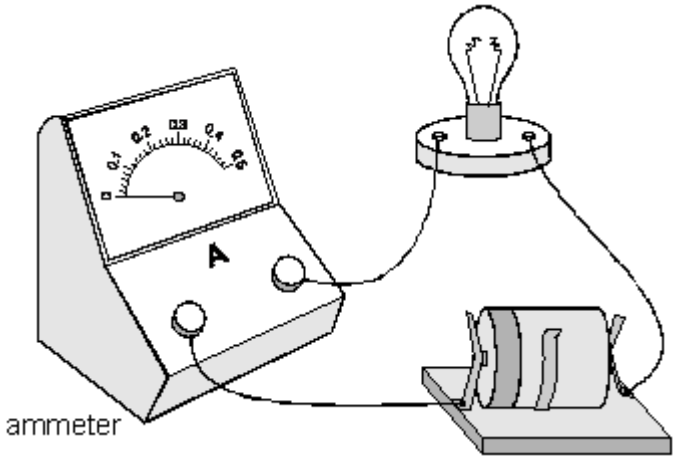


diagram 3

(c) In diagram 3 the ammeter reading is zero. Why is this **not** a complete circuit?

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1 mark

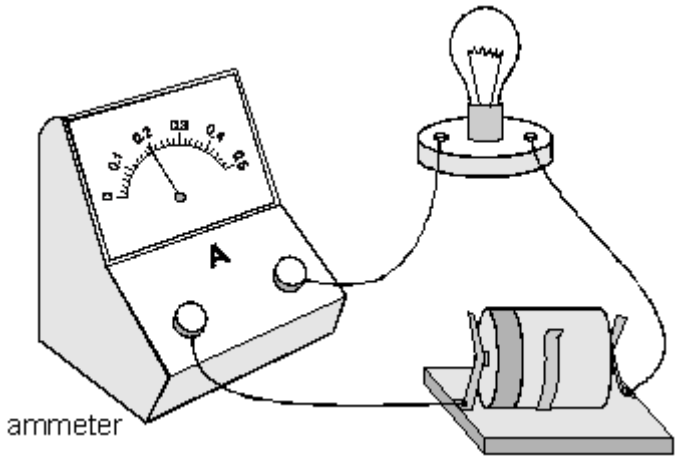


diagram 4

(d) In diagram 4, why is there a reading on the ammeter?

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1 mark

Maximum 4 marks

Mark schemes

Q1.

(a) there is a wire missing between the battery and the bulb

accept 'there is a wire missing'
or *'the bulb is not connected to the battery'*
accept 'it is not a complete circuit'
or *'the circuit is broken'*

1 (L3)

(b) two wires are connected to one end of the battery

accept 'the battery is wrongly connected'
accept 'the battery is not in the circuit'
accept 'the circuit is wrongly connected'
do ***not*** *accept 'there is an incomplete circuit'*

1 (L3)

(c) the bulb is broken **or** blown

do ***not*** *accept 'the circuit is not complete'*
do ***not*** accept *'because the light is not on'*

1 (L3)

(d) the circuit is connected correctly

accept 'it is set up right'
accept 'there is a current'
accept 'there is a complete circuit ***or*** *path'*

1 (L3)

[4]